

1 Overview

1. Macroeconomics studies aggregate variables, like aggregate consumption, aggregate investment.
2. Questions are typically dynamic, hence dynamic programming is useful.
3. Macro models with microfoundations, i.e., models where behavior is derived from basic assumptions on consumers' preferences, production technologies etc.
4. Two types of macro models: One generation where agents are infinitely lived, multiple generations where agents are finitely lived
5. Two types of timing: continuous-time v.s. discrete-time.

2 A finite horizon problem

$$\sum_{t=0}^T \beta^t u(c_t),$$

where $\beta \in (0, 1)$ is called the discount factor. β measures patience.

The optimization problem

$$\begin{aligned} \max \quad & \sum_{t=0}^T \beta^t u(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq f(k_t) \equiv F(k_t, 1) + (1 - \delta)k_t \\ & c_t \geq 0, \forall t = 0, 1, \dots, T \\ & k_{t+1} \geq 0, \forall t = 0, 1, \dots, T \\ & k_0 > 0 \text{ is given.} \end{aligned}$$

A social planner's problem: there is no market where individuals trade.

Note that

1. In each period, there is only one good (either for consumption or for investment). Potato.
2. It does not matter whether the feasibility condition is $c_t + k_{t+1} = F(k_t, 1) + (1 - \delta)k_t$ or $c_t + k_{t+1} \leq F(k_t, 1) + (1 - \delta)k_t$.

Assumption 1 *u is strictly increasing, differentiable, concave and f is strictly increasing, differentiable, concave. $F(0, L) = F(K, 0) = 0$.*

First we know that $c_t + k_{t+1} < f(k_t) + (1 - \delta)k_t$ cannot be optimal. Hence $c_t = f(k_t) - k_{t+1}$. Then the problem is

$$\begin{aligned} \max \quad & \sum_{t=0}^T \beta^t u(f(k_t) - k_{t+1}), \\ \text{s.t.} \quad & 0 \leq k_{t+1} \leq f(k_t), \forall t = 0, 1, \dots, T \\ & k_0 > 0 \text{ is given.} \end{aligned}$$

The constraint qualification condition (the gradients of binding constraints are linearly independent) is not easily verifiable, as we do not know which ones bind in general. Economist use a simple trick.

Assumption 2 (Inada condition on utility function) $\lim_{c \rightarrow 0} u'(c) = \infty$.

For example, CRRA utilities meet this condition. This condition says that zero consumption is undesirable, and hence consumption in all periods must be positive.

LEMMA 1 *Under the Inada condition on utility, the optimal solution is interior, that is, $k_{T+1} = 0$ and $0 < k_{t+1} < f(k_t)$ for all $t = 0, 1, \dots, T - 1$.*

LEMMA 1 *Under the Inada condition on utility, the optimal solution is interior, that is, $0 < k_{t+1} < f(k_t)$ for all $t = 0, 1, \dots, T - 1$.*

PROOF OF : First show that capital must be positive. Suppose not, let t be the period when capital starts to be zero, i.e., $k_t > 0$ and $k_{t+1} = 0$. Fix other capitals and only vary $x = k_{t+1}$, then the objective function is

$$\dots \beta^t u(f(k_t) - x) + \beta^{t+1} u(f(x))$$

The derivative of the above with respect to x is $-\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x)) f'(x)$, which is infinity at $x = 0$.

Now we show that consumption must be positive. Suppose not, then pick the last period when consumption is zero. That is, a period when $c_t = 0$ but $c_{t+1} > 0$. The derivative is

$$-\beta^t u'(f(k_t) - x) + \beta^{t+1} u'(f(x) - k_{t+2}) f'(x),$$

which is $-\infty$ when $x = f(k_t)$.

Q.E.D.

The summary statement of the first-order conditions is then the Euler equation:

$$u'[c_t] = \beta u'[c_{t+1}] f'(k_{t+1}); t = 0, \dots, T - 1$$

$$k_0 \text{ given}, k_{T+1} = 0;$$

1. Count the number of equations and unknowns. both are $T + 1$.
2. It is a second order difference equation, because there are two lags of capital.
3. are sufficient conditions if the objective function is concave. To see the concavity of $\sum_{t=0}^T \beta^t u(f(k_t) - k_{t+1})$, it is sufficient to show that $u(f(k_t) - k_{t+1})$ is concave

for all t , as the sum of concave functions is still concave. We have

$$\begin{aligned}\frac{\partial u}{\partial k_t} &= u'(c_t)f'(k_t), \\ \frac{\partial u}{\partial k_{t+1}} &= -u'(c_t), \\ \frac{\partial^2 u}{\partial k_t^2} &= u''(c_t)(f'(k_t))^2 + u'(c_t)f''(k_t) \leq 0, \\ \frac{\partial^2 u}{\partial k_{t+1}^2} &= u''(c_t) \leq 0, \\ \frac{\partial^2 u}{\partial k_t \partial k_{t+1}} &= -u''(c_t)f'(k_t).\end{aligned}$$

hence the hessian matrix of u is negative semi-definite, as

$$\frac{\partial^2 u}{\partial k_t^2} \frac{\partial^2 u}{\partial k_{t+1}^2} - \left(\frac{\partial^2 u}{\partial k_t \partial k_{t+1}} \right)^2 = u'(c_t)f''(k_t)u''(c_t) \geq 0.$$

Economic interpretation of the Euler equation.

Utility lost if you invest one more unit (i.e., marginal cost of saving) =
Utility increase next period per unit of increase in c * Return on the invested unit

EXAMPLE 1 *Numerical exploration.*

$$\begin{aligned} \max \quad & \sum_{t=0}^3 \beta^t \log(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq k_t^\alpha + (1 - \delta)k_t \\ & k_0 = 2 \text{ is given,} \end{aligned}$$

where $\alpha = \frac{1}{3}$, $\delta = 0.1$. The Euler equations are

$$\begin{aligned} \frac{1}{k_0^\alpha + (1 - \delta)k_0 - k_1} &= \frac{\beta(\alpha k_1^{\alpha-1} + 1 - \delta)}{k_1^\alpha + (1 - \delta)k_1 - k_2} \\ \frac{1}{k_1^\alpha + (1 - \delta)k_1 - k_2} &= \frac{\beta(\alpha k_2^{\alpha-1} + 1 - \delta)}{k_2^\alpha + (1 - \delta)k_2 - k_3} \\ \frac{1}{k_2^\alpha + (1 - \delta)k_2 - k_3} &= \frac{\beta(\alpha k_3^{\alpha-1} + 1 - \delta)}{k_3^\alpha + (1 - \delta)k_3 - k_4} \\ & k_4 = 0. \end{aligned}$$

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PROGRAM shooting
REAL,DIMENSION(0:4):: k
REAL:: delta=0.1,alpha=1.0/3.0,beta=0.9,R
REAL:: guess(100),step
INTEGER :: i,j
k(0)=2.0; ! initial condition
step=(k(0)**alpha+(1-delta)*k(0)-0.0001)/(SIZE(guess)-1);

! initialize the vector of guess
DO i=1,SIZE(guess)
    guess(i)=0.0001+(i-1)*step
ENDDO

!-----
! follow the difference equation (Euler equation) in a growth model

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!-----
open(1,file='shooting.csv')

DO j=1,SIZE(guess)
    k(1)=guess(j);
    DO i=2,4
        IF ((k(i-1)>0).and.(k(i-2)>0)) THEN
            R=alpha*k(i-1)**(alpha-1)+1.0-delta;
            k(i)=k(i-1)**alpha+(1-delta)*k(i-1)-beta*R*(k(i-2)**alpha+(1-delta)*k(i-2)-k(i-1));
        ELSE
            k(i)=-10000.0
        ENDIF
    ENDDO
    write(1,"(10(f20.4,' ',''))" k
ENDDO

CLOSE(1);
END PROGRAM shooting

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EXAMPLE 2 *Analytical exploration.*

$$\begin{aligned}
 \max \quad & \sum_{t=0}^T \beta^t \log(c_t), \\
 \text{s.t.} \quad & c_t + k_{t+1} \leq Rk_t \\
 & k_0 > 0 \text{ is given.}
 \end{aligned}$$

Method 1: Solve the linear difference equation,

$$\frac{1}{Rk_0 - k_1} = \frac{\beta R}{Rk_1 - k_2}$$

The two roots of the characterization equation are $\lambda_1 = R$ and $\lambda_2 = \beta R$. Hence the solution takes the form of

$$k_t = AR^t + B(\beta R)^t,$$

where A and B are undetermined parameters. $A = -\frac{\beta^{T+1}k_0}{1-\beta^{T+1}}$, $B = \frac{k_0}{1-\beta^{T+1}}$. Hence

$$\begin{aligned} c_t &= Rk_t - k_{t+1} = [R(AR^t + B(\beta R)^t) - AR^{t+1} - B(\beta R)^{t+1}]k_0 \\ &= \frac{(1-\beta)\beta^t R^{t+1}}{1-\beta^{T+1}}k_0 \end{aligned}$$

Method 2: Consolidate the budget constraints and then solve for the multiplier conditions.

$$c_0 + R^{-1}c_1 + \dots + R^{-T}c_T = Rk_0$$

The first-order condition is

$$-\frac{\beta^t}{c_t} + \lambda R^{-t} = 0$$

Hence $c_t = \frac{(\beta R)^t}{\lambda}$. We have

$$\lambda = \frac{1 + \beta + \beta^2 + \dots + \beta^T}{Rk_0},$$

thus $c_t = \frac{Rk_0(\beta R)^t}{1 + \beta + \beta^2 + \dots + \beta^T}$.

2.1 Equilibrium with sequential trade

What allocation will be achieved in decentralized trade?

Let us focus on the case with constant-returns-to-scale production function.

1. Real return of saving: R_t .
2. Price of labor: w_t : wage rate; price of labor at t relative to (in terms of) consumption goods at t :

A competitive equilibrium is a sequence of prices and allocations, such that

1. All the decision makers maximize their utility or profit.
2. In every market, supply=demand.

Allocations are $\{c_t^*, K_t^{s*}, K_t^{d*}, n_t^{s*}, n_t^{d*}\}_{t=0}^T$ and prices are $\{R_t^*, w_t^*\}_{t=0}^T$.

1.

$$\begin{aligned} \{c_t^*, K_t^{s*}, n_t^{s*}\}_{t=0}^T &\in \arg \max \sum_{t=0}^T \beta^t u(c_t) \\ \text{s.t.} \quad &c_t + K_{t+1} = R_t^* K_t + w_t^* n_t \\ &K_0^{s*} = K_0 \text{ is given} \end{aligned}$$

2.

$$K_t^{d*}, n_t^{d*} \in \arg \max_{K, n} F(K, n) + (1 - \delta)K - w_t^* n - R_t^* K$$

3. $n_t^{s*} = n_t^{d*}$, for all t .

4. $K_t^{s*} = K_t^{d*}$, for all t .

REMARK 1 *The economy has a continuum of individuals who are identical. Hence this is a representative-agent economy.*

REMARK 2 *The number of firms is undetermined, as long as we have many small firms and the price-taking behavior can be justified. The economy has a continuum of individuals who are identical. Hence this is a representative-agent economy.*

REMARK 3 *Interpretation of the goods market.*

REMARK 4 *This is a perfect foresight model: individuals know what the future prices would be when making their decisions in period zero.*

Let us analyze the conditions that characterize an equilibrium

1. $1 = n_t^{s*} = n_t^{d*}$.
2. First-order condition for a consumer

$$\begin{aligned}\beta^t u'(c_t^*) &= \lambda_t \\ -\lambda_t + R_{t+1}^* \lambda_{t+1} &= 0.\end{aligned}$$

hence we have

$$\beta^t u'(c_t^*) = \beta^{t+1} u'(c_t^*) R_{t+1}^*.$$

3. First-order condition for the firm $R_t^* = F_K(K_t^*, 1) + (1 - \delta)$ and $w_t^* = F_N(K_t^*, 1)$.
4. CRS implies that $F(K_t, n_t) + (1 - \delta)K_t = R_t^* K_t + w_t^* n_t$. Firm's profit is zero.
5. That the consumer's budget constraint is satisfied implies that the feasibility condition in the social planner's problem is satisfied.
6. Euler equation in the social planner's problem holds.
7. $K_{T+1} = 0$.

THEOREM 1 (FIRST WELFARE THEOREM) *The allocation in a competitive equilibrium solves the social planner's problem (it is pareto efficient).*

REMARK 5 *Because there is a difference between the aggregate variable and an individual variable, the consumer's budget constraint is not the same as the feasibility condition in the social planner's problem.*

EXAMPLE 3 *Suppose $u(c) = \log(c)$ and $f(k) = k^\alpha$ and $T = 1$, $k_0 = 1.0$, $\alpha = 1/2$, $\delta = 1$, $\beta = 2/3$.*

- *The social planner's feasibility set is $\{(c_0, c_1) : c_1 \leq (k_0^\alpha - c_0)^\alpha\}$...*
- *The solution is $c_0 = \frac{1}{1+\alpha\beta}k_0^\alpha = \frac{3}{4}$, $k_1 = \frac{\alpha\beta}{1+\alpha\beta}k_0^\alpha = \frac{1}{4}$, $c_1 = k_1^\alpha = \frac{1}{2}$.*
- *The prices are $R_1 = \alpha k_1^{\alpha-1} = 1$, $w_1 = (1 - \alpha)k_1^\alpha = \frac{1}{4}$.*
- *The consumer's budget set is a triangle $\{(c_0, c_1) : c_0 + c_1 \leq 1 + w_1\}$.*

THEOREM 2 (SECOND WELFARE THEOREM) *If an allocation is pareto efficient, then we can find prices such that the prices and the allocation constitute an equilibrium.*

This is for students who did not attend the summer math course.

1. Fortran program
2. Upper case and lower case letters are not distinguished.
3. Four data types
 - (a) REAL, or REAL*8
 - (b) INTEGER,
 - (c) LOGICAL: .T. .F.
 - (d) CHARACTER
4. Operators: +,-,*,/,**
5. Build-in functions: EXP, LOG, SIN, COS, MATMUL, DOT_PRODUCT. There is no function to compute the inverse of a matrix.
6. IF THEN; ELSE; ENDIF
7. DO i=1,100
ENDDO
8. Functions and Subroutines
9. Newton's method
 - (a) $f(x) = 2 \sin(x) - x = 0$
 - (b)

$$\begin{aligned}\frac{1}{3}(x_2) + (x_1)^2(x_2) - \frac{2}{3}(x_1)^2 &= 0, \\ \frac{2}{3}(x_1) - \frac{1}{3}(x_2)^2 - 1 &= 0.\end{aligned}$$